Advanced AI (Section A)

Assignment Report

Image Segmentation with UNet and UNet++, and an Evaluation of Cityscapes-Trained Models on Karachi Road Scenes

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Code, notebooks and all result files: https://github.com/KiranKhann/karachi-road-segmentation

1. Introduction

This assignment had four parts: to summarise the two assigned papers, to prepare a notebook that implements UNet and UNet++ in a way that can be reused for any segmentation problem, to study how the Cityscapes dataset represents its fifty cities and then segment the objects in busy Karachi road scenes while respecting privacy, and finally to review the segmentation work presented at CVPR 2025 and ICML 2024. This report describes what was done for each part, the results obtained, and what I learned from them.

All experiments were run on a laptop with an NVIDIA RTX 4050 (6 GB) GPU using Python 3.14, PyTorch 2.14 and the Hugging Face Transformers library. Two Jupyter notebooks were produced: *01_unet_unetpp_framework.ipynb* (UNet and UNet++ written from scratch with a reusable training framework) and *02_karachi_road_segmentation.ipynb* (segmentation of Karachi road photographs with two Cityscapes-trained networks). Both notebooks were executed end to end and are included, together with their outputs, in the project repository.

2. Summary of the Assigned Papers

2.1 Abramova et al. (2021): Hemorrhagic stroke lesion segmentation using a 3D U-Net with squeeze-and-excitation blocks

Hemorrhagic stroke is bleeding inside the brain. It has a high mortality rate, and the volume and position of the bleed on a non-contrast CT scan are important for treatment, so the authors set out to segment these lesions automatically instead of relying on slow manual outlining [3]. Their network is a 3D U-Net, i.e. the standard encoder-decoder with skip connections applied to whole volumes, in which every convolutional block is followed by a squeeze-and-excitation (SE) block [4]. The SE block pools each feature channel to a single number, passes the vector through two small fully connected layers and multiplies the result back onto the channels, so the network learns which channels to emphasise for the lesion.

The training procedure is as interesting as the architecture. Because a bleed occupies a tiny fraction of the scan, the authors use a restrictive patch sampling scheme that only draws training patches from a region of interest; this balances the classes and also keeps intra-ventricular haemorrhage, which they do not count as a stroke lesion, out of the training signal. They further add the mid-sagittally flipped scan as an extra input channel (symmetric modality augmentation) so that the network can compare the two hemispheres in the same way a radiologist does, and they ablate the patch size, input modalities, augmentation and loss functions. With five-fold cross-validation on 76 clinical cases the full method reached a mean Dice coefficient of 0.86 ± 0.07, and the SE blocks gave a consistent improvement over the plain 3D U-Net.

The main lesson I took from this paper is that in medical segmentation the gains usually come from small, well-motivated additions to the UNet design (attention, better sampling, a better loss) rather than from a new backbone. For that reason the UNet implementation in Task 2 includes an optional SE block that reproduces the same idea.

2.2 Kirillov et al. (2019): Panoptic Segmentation

Semantic segmentation assigns a class label to each pixel but cannot separate two neighbouring cars; instance segmentation detects and masks individual objects but ignores amorphous regions such as road and sky. Kirillov et al. propose panoptic segmentation as a single task that does both [5]: every pixel receives a class label and, for countable ("thing") classes, an instance identifier, and the segments may not overlap. The paper also introduces the panoptic quality (PQ) metric. Predicted and ground-truth segments are matched when their IoU exceeds 0.5, which makes the matching unique, and PQ is the product of segmentation quality (the mean IoU of matched segments) and recognition quality (an F1-style score of matched, missed and spurious segments). PQ treats stuff and thing classes in exactly the same way.

The authors measured PQ for humans and for a simple machine baseline that merges the outputs of an instance model (Mask R-CNN) and a semantic model (PSPNet) on Cityscapes, ADE20k and Mapillary Vistas. Humans reached roughly 70 PQ on Cityscapes and 57 on Vistas, far above the 2018 baselines. The paper succeeded in its aim of reviving interest in unified scene segmentation: Cityscapes and COCO now have panoptic benchmarks, and unified architectures such as Panoptic-FPN, MaskFormer and Mask2Former [8] followed. The Mask2Former model used in Task 3 is a direct descendant of this work.

2.3 Background: UNet and UNet++

UNet [1] is a symmetric encoder-decoder network in which feature maps from each encoder level are copied to the decoder level of the same resolution. These skip connections restore the detail lost by pooling and allow the network to be trained from few images with strong augmentation, which is why it became the default for biomedical segmentation. UNet++ [2] keeps the encoder but replaces the plain skips with nested, densely connected convolution blocks. A decoder node at depth i and column j receives all earlier nodes at the same depth plus the upsampled node from one level deeper, so encoder features are refined step by step before they are fused with the decoder. UNet++ also supervises all four nested outputs (deep supervision), which allows the trained network to be pruned at inference time to a shallower, faster sub-network.

3. Task 2: A Reusable UNet / UNet++ Notebook

3.1 Design

The notebook is organised so that a new use case only requires editing one configuration cell. The *Config* dataclass holds the image and mask folders, the number of classes (1 for a binary mask, K for a multi-class index mask), the training resolution, the number of base channels, the epochs, learning rate and the two architectural switches (*use_se* for SE blocks and *deep_supervision* for UNet++). A dataset class pairs images with masks by file name, resizes them (bilinear for images, nearest-neighbour for masks), caches them in memory and applies flips, 90-degree rotations and brightness jitter during training. UNet and UNet++ are implemented from scratch in PyTorch; the UNet++ forward pass takes an argument L in 1 to 4 that selects how many nested columns to evaluate, which is how pruning is realised. The loss is binary cross-entropy (or cross-entropy for several classes) plus a soft Dice term, and the trainer uses AdamW, a one-cycle learning-rate schedule, mixed precision, early stopping on the

validation Dice and checkpointing. The notebook ends with a comparison section, a prediction function that works at any image resolution and a short checklist for adapting it to other data.

3.2 Dataset and training setup

As a worked example I used Kvasir-SEG [9], a public set of 1,000 colonoscopy frames with polyp masks (CC BY 4.0), which the notebook downloads automatically; if the download fails it falls back to a synthetic shapes dataset so that every cell still runs. The data were split 850/150 for training and validation, resized to 256 × 256 and trained for 12 epochs with batch size 16 and base width 32. One epoch took about 15 s for UNet and 52 s for UNet++ on the RTX 4050.

3.3 Results

Table 1: UNet and UNet++ on the Kvasir-SEG validation split after 12 epochs (same split, same schedule).

Model	Parameters	Val. loss	Val. Dice	Val. IoU	Latency (ms, 256^2)
UNet	7.85 M	0.530	0.735	0.629	5.4
UNet++ pruned, L = 1	9.16 M (shared)	1.006	0.389	0.268	2.2
UNet++ pruned, L = 2	9.16 M (shared)	0.842	0.532	0.393	5.1
UNet++ pruned, L = 3	9.16 M (shared)	0.613	0.695	0.570	10.3
UNet++ full, L = 4	9.16 M	0.497	0.765	0.663	16.4

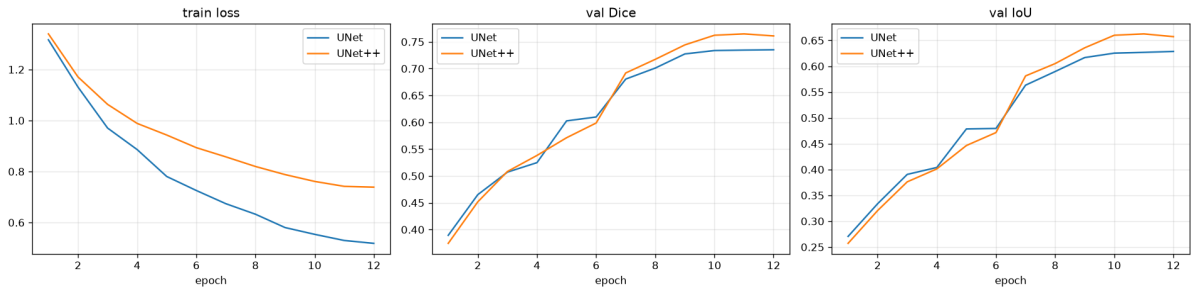


Figure 1: Training loss, validation Dice and validation IoU per epoch for UNet and UNet++.

UNet++ finished 3.0 Dice points and 3.4 IoU points above UNet, which agrees in direction and roughly in size with the improvements reported by Zhou et al. [2], at about three times the inference cost. The pruning rows reproduce the second claim of that paper: the deep-supervised network can be cut at inference. Stopping at L = 3 keeps 91% of the full Dice at 63% of the latency, while L = 2 matches the speed of UNet but, after only 12 epochs, not its accuracy. Both validation curves in Figure 1 are still rising at the last epoch, so these numbers are a short demonstration rather than a converged benchmark; published results on Kvasir-SEG with pretrained encoders and long schedules reach Dice above 0.90. Figure 2 shows six validation cases. Both networks handle large, well-defined polyps well; the failures are small or flat polyps and frames with strong specular highlights, where UNet++ tends to produce fewer spurious regions than UNet.

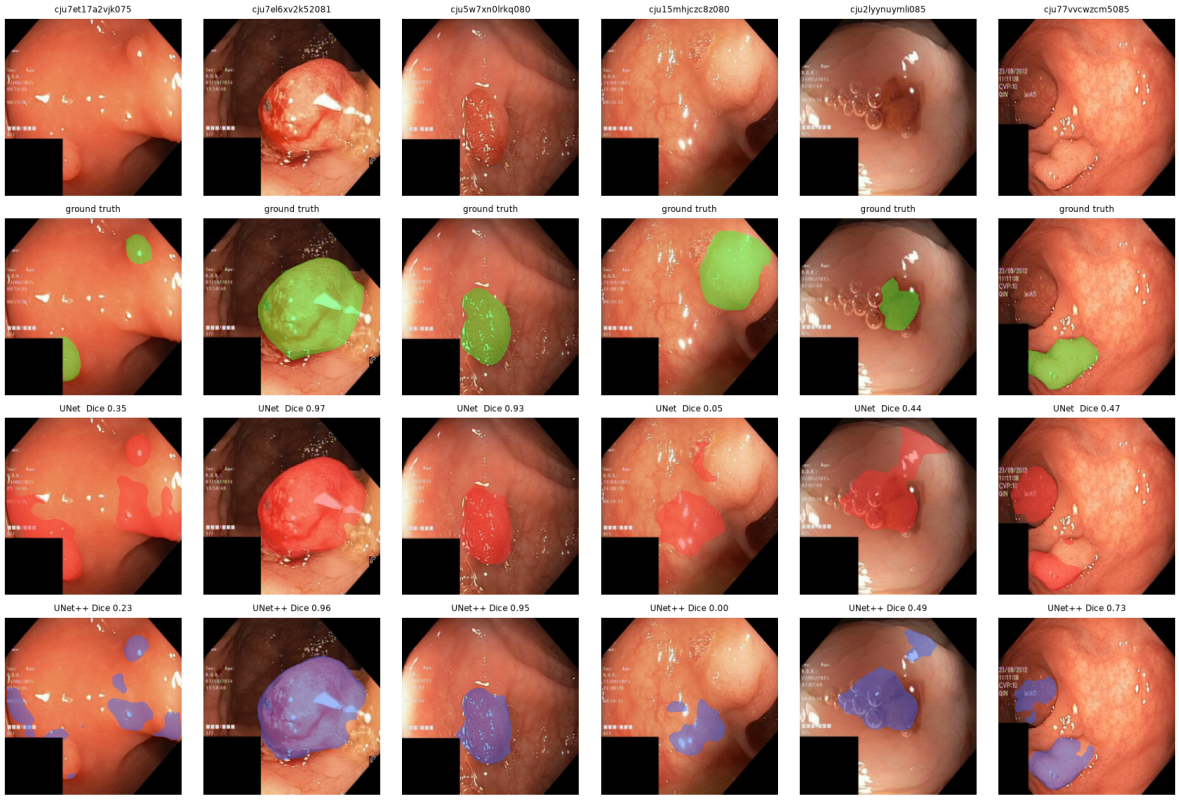


Figure 2: Validation examples: input, ground truth (green), UNet prediction (red) and UNet++ prediction (blue) with per-image Dice.

4. Task 3: Cityscapes and Segmentation of Karachi Road Scenes

4.1 How Cityscapes represents fifty cities

Cityscapes [6] was recorded from a car in 50 cities, almost all of them in Germany, with Zürich (Switzerland) and Strasbourg (France) as the only exceptions. Only 27 cities have the fine, pixel-accurate annotations (training: Aachen, Bochum, Bremen, Cologne, Darmstadt, Düsseldorf, Erfurt, Hamburg, Hanover, Jena, Krefeld, Mönchengladbach, Strasbourg, Stuttgart, Tübingen, Ulm, Weimar and Zürich; validation: Frankfurt, Lindau and Münster; test: Berlin, Bielefeld, Bonn, Leverkusen, Mainz and Munich). The other 23 cities, for example Saarbrücken, Nuremberg, Erlangen, Bamberg and Dortmund shown on the examples page, appear only in the coarsely annotated set. There are 5,000 fine and 20,000 coarse frames at 2048 × 1024 pixels; each annotated frame is the 20th frame of a 30-frame stereo clip and comes with GPS, odometry and temperature readings. All recordings were made in daytime, between spring and autumn and in good or medium weather, so there is no night, rain or snow. The label set has 30 classes in eight groups (Table 2); 19 of them are used for evaluation and eight of those carry instance masks. The website notes that the released videos are blurred for privacy.

Table 2: The Cityscapes label hierarchy.

Group	Classes (+ = not evaluated, * = instance labels)
flat	road, sidewalk, parking+, rail track+
human	person*, rider*
vehicle	car*, truck*, bus*, on rails*, motorcycle*, bicycle*, caravan*+, trailer*+
construction	building, wall, fence, guard rail+, bridge+, tunnel+
object	pole, pole group+, traffic sign, traffic light
nature	vegetation, terrain
sky	sky
void	ground+, dynamic+, static+

What struck me is how consistent the dataset is: one camera rig, one viewpoint, one set of traffic rules and similar architecture everywhere. This consistency is why Cityscapes models exceed 80% mean IoU on its own test set, but it also means the models learn a strictly Central-European idea of a street: right-hand traffic, lane markings, kerbed sidewalks, cycle lanes, sedans and hatchbacks, and very few pedestrians on the carriageway. Karachi differs on nearly every one of these points, which makes it a good stress test for such models.

4.2 Data collection and ethical considerations

Photographing people in Karachi traffic for a course project raises consent and safety concerns, so instead of taking new pictures I used 22 openly licensed photographs of Karachi roads from Wikimedia Commons (CC BY, CC BY-SA and Free Art licences), covering Saddar, M. A. Jinnah Road, Shahrah-e-Faisal, I. I. Chundrigar Road, the Nagan flyover, Federal B Area, Shaheed-e-Millat Road and other locations, including one night scene. The author, licence, date and source page of every file are stored in a manifest and shown with the figures, as the licences require.

Two anonymisation passes are applied before any image is displayed or saved. First, faces and number plates are detected with the YuNet face detector and the LPD-YuNet plate detector from the OpenCV Model Zoo [12] and replaced by a coarse mosaic before the segmentation models see the image. Second, after panoptic segmentation, every pixel of every detected person or rider is mosaiced (a face detector can miss turned or partly hidden faces in a crowd, but a segmentation mask does not), and the lower band of every car, truck and bus instance, where the plate sits, is mosaiced as well. Over the 22 images this removed 50 detected faces, 5 detected plates, 241 person segments and 75 vehicle plate zones. Only the resulting images (Figure 3) are used in figures, and the analysis itself produces aggregate statistics only; no tracking or identification is attempted.



Figure 3: The 22 Karachi road scenes after anonymisation, with photographer and licence.

4.3 Models

Two publicly available networks trained on Cityscapes were used. SegFormer-B2 [7] (27 M parameters) is a hierarchical Transformer encoder with a light MLP decoder; its logits were upsampled to the image size and both the arg-max label and the maximum softmax probability were kept, the latter as a cheap confidence signal. Mask2Former with a Swin-T backbone [8] (47 M parameters) was used in its panoptic configuration, so that cars, motorbikes, buses and people could be counted per image. The default input sizes of the Hugging Face processors (512 and 384 pixels) are too small for street scenes, so they were raised to 1024 pixels for SegFormer and an 800-pixel short side for Mask2Former; inference over all images took about 18 s with a peak of 2.2 GB of GPU memory.

4.4 Results



Figure 4: SegFormer-B2 predictions on six of the Karachi scenes.

Table 3: Summary statistics of the Karachi experiment.

Measure	Value
Mean SegFormer confidence / pixels below 0.5	0.945 / 2.0%
Pixel agreement between SegFormer and Mask2Former	83% on average (minimum 56% on the Saddar street-market image); correlation with confidence 0.54
Mean pixel composition	building 22.5%, road 22.2%, sky 19.6%, vegetation 10.9%, car 8.5%
Instances counted over all images (Mask2Former)	213 car, 197 person, 44 rider, 36 bicycle, 11 motorcycle, 3 truck, 2 bus
Most / least confident classes	sky, road, building / fence, terrain, wall, traffic sign, pole
Two-wheelers (motorcycle + bicycle labels)	18% of car plus two-wheeler instances

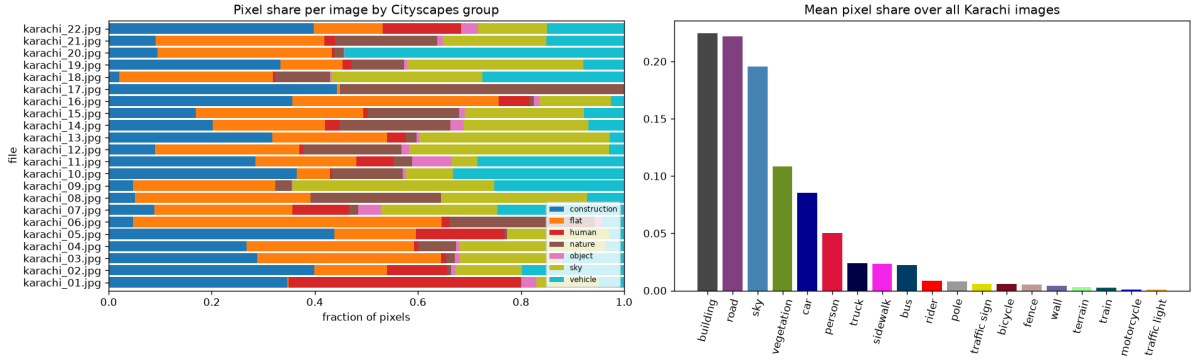


Figure 5: Left: pixel share per image by Cityscapes group. Right: mean pixel share per class over all images.

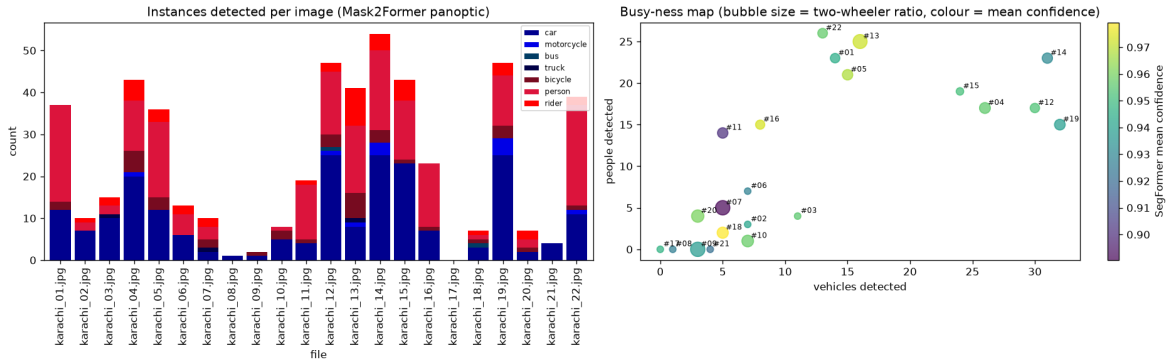


Figure 6: Left: instances detected per image. Right: vehicles versus people per image; bubble size is the two-wheeler ratio and colour the mean SegFormer confidence.

The large stuff classes transfer well. Road, building, sky and vegetation are segmented cleanly and with high confidence, the two models agree on 83% of pixels, and agreement rises with SegFormer's confidence, so confidence is a usable quality signal even without Karachi labels. Cars and people are also found reliably in dense scenes, and the instance counts give a believable picture of how busy each road is (Figure 6).

The failures are concentrated in exactly the places where Karachi differs from a German city. The clearest case is the two-wheelers: Mask2Former returned three times more "bicycle" than "motorcycle" instances, although Karachi has very few bicycles. Figure 7 shows the largest instances of each label. The "bicycles" are motorbikes with riders, rows of parked motorbikes merged into one segment, a hand-cart and even a rickshaw, while a third of the "motorcycles" are auto-rickshaws. The two-wheeler share of 18% in Table 3 is therefore an under-estimate, and any traffic statistic built on the raw class names would mis-count the vehicle types that dominate Karachi traffic.

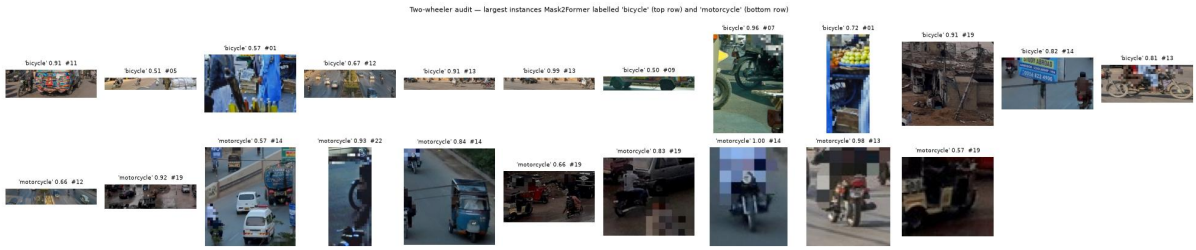


Figure 7: The largest instances labelled bicycle (top) and motorcycle (bottom) by Mask2Former.

Objects that have no Cityscapes class, such as rickshaws, qingqis, hand-carts and hawkers' goods, are forced into the nearest available label and are where the lowest-confidence regions occur (Figure 9): the canopy of a rickshaw was labelled sky with a confidence of 0.30, and the body of a decorated bus received 0.35. Decorated buses in general were split between bus, truck, train and building (only two bus instances were counted although at least five buses are visible), Urdu signboards on building facades produced the largest disagreement between the two models (41% of SegFormer's traffic-sign pixels are building for Mask2Former, Figure 10), and the single night image produced no instances at all, which is unsurprising since Cityscapes contains no night data. Finally, the absence of kerbed sidewalks means that a quarter of SegFormer's sidewalk pixels are called road or terrain by Mask2Former, and Suzuki pick-ups and vans sit between the truck and car classes (36% disagreement).

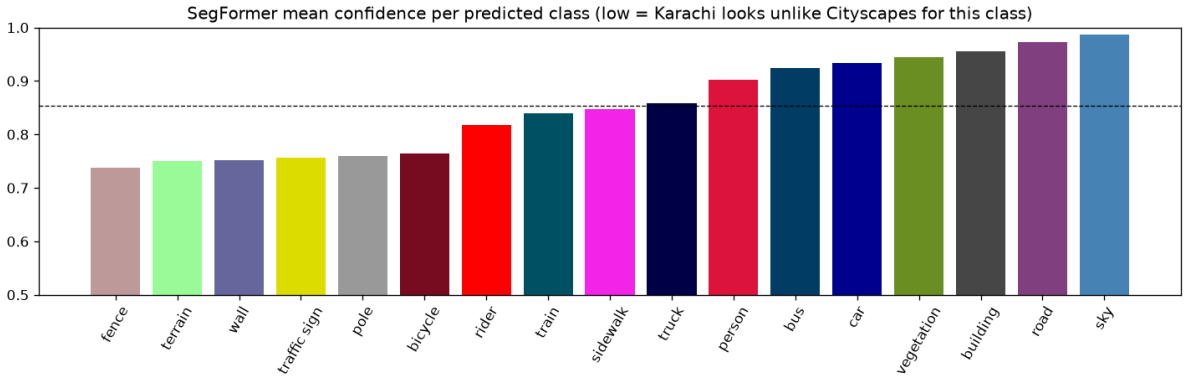


Figure 8: Mean SegFormer confidence per predicted class over all Karachi images.



Figure 9: Lowest-confidence regions and the label SegFormer assigned to them.

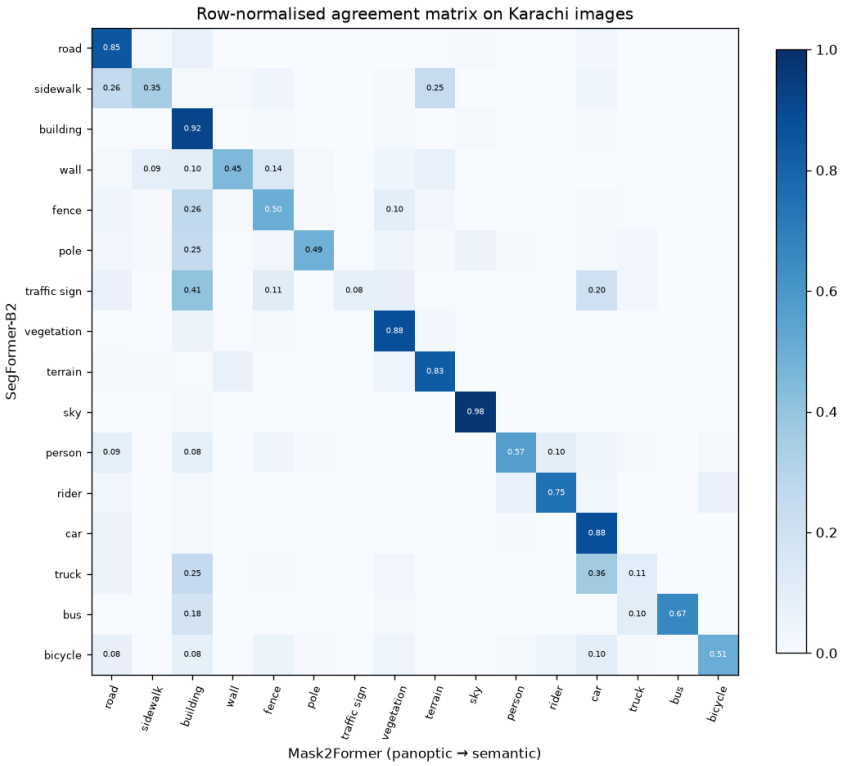


Figure 10: Row-normalised agreement between SegFormer (rows) and Mask2Former (columns) on the Karachi images.

4.5 Discussion

A model trained only on Cityscapes gives a good description of the layout of a Karachi street but unreliable object statistics. The practical remedy is local data. The notebook therefore exports pseudo-labels (pixels on which both models agree with confidence above 0.7, about

80% of all pixels) so that the UNet++ from Task 2 or SegFormer itself can be fine-tuned on Karachi images, and the India Driving Dataset [10], which adds classes such as autorickshaw, animal and drivable fallback, shows what a Karachi label set should contain. For the Pakistani context the CARL-D dataset [13], recorded in more than 100 Pakistani cities with pixel-level labels for rickshaws, carts and unpaved roads, would be the natural next training set.

5. Task 4: State-of-the-Art Segmentation at CVPR 2025 and ICML 2024

For this part I went through the CVPR 2025 accepted-paper list, the oral and award pages and the ICML 2024 call for papers, proceedings index and award page. Only papers that appear on those pages are cited here.

5.1 CVPR 2025 (Nashville, 11 to 15 June 2025)

Table 4: Main segmentation trends at CVPR 2025.

Trend	What changed	Example papers at CVPR 2025
Promptable foundation models (SAM, SAM 2)	Segment Anything is the starting point; work extends it to occlusion, video, medical images and on-device use	Segment Anything, Even Occluded; Enhancing SAM with Efficient Prompting and Preference Optimization for Semi-supervised Medical Image Segmentation (oral); EntitySAM; EdgeTAM
Open-vocabulary segmentation	Arbitrary text queries instead of a fixed label set, usually built on CLIP or vision-language models	Exploring Simple Open-Vocabulary Semantic Segmentation; Dual Semantic Guidance for Open Vocabulary Semantic Segmentation; v-CLR (highlight)
Training-free / frozen backbones	Frozen CLIP, DINO or diffusion features reused with little or no training	SegEarth-OV (oral); ResCLIP; Advancing Generalizable Tumor Segmentation with Frozen Foundation Diffusion Models
Video and reasoning segmentation	Temporal consistency, referring expressions and language-driven reasoning about what to segment	The Devil is in Temporal Token; Semantic and Sequential Alignment for Referring Video Object Segmentation; ViCaS; RipVIS
3D: LiDAR and Gaussian splatting	Segmentation inside 3D representations, often language-grounded	PanoGS; iSegMan; Dr. Splat (highlight); No Thing, Nothing (LiDAR in adverse weather); STU anomaly dataset
Medical and unified architectures	Universal medical segmenters, in-context learning, Mamba backbones, one model for many datasets	Show and Segment; Cross-Modal Interactive Perception Network with Mamba; Scaling up Image Segmentation across Data and Tasks; Mamba as a Bridge (highlight)

None of the CVPR 2025 award papers is about segmentation: the best paper was VGGT (Visual Geometry Grounded Transformer) [16], the best student paper was on neural inverse rendering, and the honourable mentions were MegaSaM, Navigation World Models, Molmo and PixMo, and 3D Student Splatting and Scooping. This itself says something about the field: the centre of gravity has moved from task-specific segmentation architectures such as DeepLab, PSPNet and Mask2Former towards general geometry and multimodal foundation models, with segmentation increasingly obtained from them (SAM 2 [18], vision-language models, 3D Gaussian splatting) rather than trained separately. For the Karachi problem the

open-vocabulary and training-free methods are particularly relevant, because a CLIP- or SAM-based segmenter can be asked for "rickshaw" without any Pakistani training data, and the domain-generalisation line of work addresses the Europe-to-South-Asia shift measured in Task 3.

5.2 ICML 2024 (Vienna, 21 to 27 July 2024)

ICML is a general machine-learning conference; its call for papers lists no vision or segmentation track, so the segmentation papers there are about learning methodology rather than new segmenters. The main themes I found in the PMLR volume 235 proceedings are listed in Table 5.

Table 5: Segmentation-related themes at ICML 2024.

Theme	Papers at ICML 2024
Adapting Segment Anything with little data	BLO-SAM: Bi-level Optimization Based Finetuning of the Segment Anything Model for Overfitting-Preventing Semantic Segmentation [14]. The SAM weights and a learned prompt embedding are optimised on two disjoint data splits at two levels, so SAM segments automatically without clicks and does not overfit on small medical sets.
Label-efficient segmentation	Active Label Correction for Semantic Segmentation with Foundation Models [15]; Hyperbolic Active Learning for Semantic Segmentation under Domain Shift; Scribble-Supervised Semantic Segmentation with Prototype-based Feature Augmentation
Robustness and certification of dense prediction	CosPGD: an efficient white-box adversarial attack for pixel-wise prediction tasks; BadPart: Unified Black-box Adversarial Patch Attacks against Pixel-wise Regression Tasks; Adaptive Hierarchical Certification for Segmentation using Randomized Smoothing
Architecture and representation	GeminiFusion: Efficient Pixel-wise Multimodal Fusion for Vision Transformer; Discrete Latent Perspective Learning for Segmentation and Detection; PruNeRF: Segment-Centric Dataset Pruning via 3D Spatial Consistency

The ten ICML 2024 best papers (among them Stealing Part of a Production Language Model, Genie, Scaling Rectified Flow Transformers and VideoPoet) contain no segmentation work, and the test-of-time award went to DeCAF, the 2014 paper that showed ImageNet features transfer to other vision tasks, which in hindsight is the ancestor of today's foundation-model segmentation.

5.3 Summary

On the leaderboards the state of the art is still held by large mask-classification Transformers (Mask2Former- and OneFormer-style decoders on ViT, Swin or InternImage backbones, often pretrained with DINOv2 or EVA-CLIP), with more than 85% mean IoU on Cityscapes and more than 58 PQ on COCO panoptic. The research frontier, however, is (i) promptable and open-vocabulary foundation models that remove the fixed label set, (ii) label-efficient adaptation of those models, (iii) video, 3D and Gaussian-splatting segmentation, and (iv) trustworthiness, meaning uncertainty, certification and robustness to domain shift, which is precisely the failure mode exposed by the Karachi experiment. UNet-family networks remain the practical state of the art for small-data 2D and 3D problems such as the stroke CT task of

the first assigned paper.

6. Conclusion

The assignment produced a reusable UNet / UNet++ framework that reproduced the main claims of the UNet++ paper on a public polyp dataset (a 3-point Dice gain over UNet and a working pruning trade-off), and an evaluation of two Cityscapes-trained networks on Karachi roads that quantified where a European street-scene prior breaks down: the large background classes transfer, but two- and three-wheelers, decorated buses, Urdu signage, unkerbed roads and night scenes do not. The privacy pipeline (detector-based face and plate blurring followed by mask-based mosaicing of every person and plate zone) worked without manual intervention and is, in my opinion, the minimum that should accompany any street-level dataset. The literature review suggests that the next step for Karachi is not a new architecture but local data: a Pakistani dataset such as CARL-D or a small, consented, anonymised Karachi set used to fine-tune UNet++ or SegFormer, possibly combined with an open-vocabulary model to recover the missing rickshaw and cart classes.

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[16] J. Wang et al., "VGGT: Visual Geometry Grounded Transformer," in *CVPR*, 2025 (best paper).

[17] ICML 2024 conference pages: https://icml.cc/Conferences/2024/CallForPapers and https://proceedings.mlr.press/v235/

[18] N. Ravi et al., "SAM 2: Segment anything in images and videos," arXiv:2408.00714, 2024.

[19] Wikimedia Commons photographs of Karachi used in Task 3; individual authors and licences are listed in data/karachi_raw/MANIFEST.csv of the project repository.